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CoPS: Conditional Prompt Synthesis for Zero-Shot Anomaly Detection

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Abstract

Recently, large pre-trained vision-language models have shown remarkable performance in zero-shot anomaly detection (ZSAD). With fine-tuning on a single auxiliary dataset, the model enables cross-category anomaly detection on diverse datasets covering industrial defects and medical lesions. Compared to manually designed prompts, prompt learning eliminates the need for expert knowledge and trial-and-error. However, it still faces the following challenges: (i) static learnable tokens struggle to capture the continuous and diverse patterns of normal and anomalous states, limiting generalization to unseen categories; (ii) fixed textual labels provide overly sparse category information, making the model prone to overfitting to a specific semantic subspace. To address these issues, we propose Conditional Prompt Synthesis (CoPS), a novel framework that synthesizes dynamic prompts conditioned on visual features to enhance ZSAD performance. Specifically, we extract representative normal and anomaly prototypes from fine-grained patch features and explicitly inject them into prompts, enabling adaptive state modeling. Given the sparsity of class labels, we leverage a variational autoencoder to model semantic image features and implicitly fuse varied class tokens into prompts. Additionally, integrated with our spatially-aware alignment mechanism, extensive experiments demonstrate that CoPS surpasses state-of-the-art methods by 1.4% in classification AUROC and 1.9% in segmentation AUROC across 13 industrial and medical datasets. The code is available at https://github.com/cqylunlun/CoPS.

1. Introduction

Anomaly detection (AD) aims to identify outliers that deviate from the normal data distribution, including industrial defect detection [5, 13, 40] and medical lesion localization [16, 27, 36]. Classical AD methods [26, 33, 41] are typically based on unsupervised learning using only normal samples and are trained to localize anomalies within known categories. However, these methods are restricted to the categories seen during training and generalize poorly to novel classes with distribution shifts during testing. This limitation poses significant challenges in real-world scenarios, where collecting sufficient data across all categories is often impractical, thereby motivating the need for zero-shot learning approaches.

To tackle this challenge, zero-shot anomaly detection (ZSAD) has been proposed to directly detect anomalies in previously unseen categories by leveraging single auxiliary training data from disjoint classes. Recently, several methods [4, 8, 9, 17, 38, 46] have demonstrated promising ZSAD

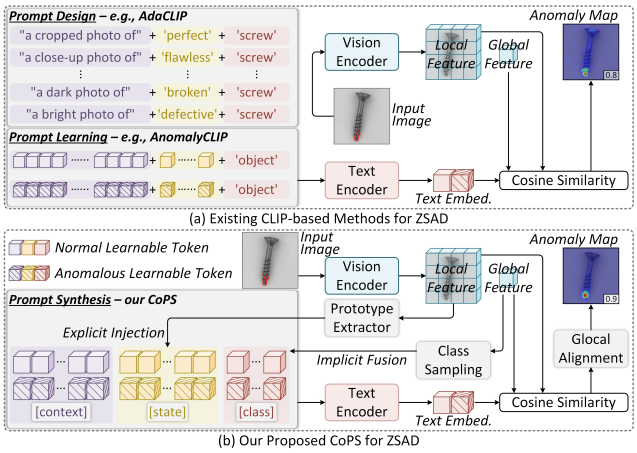


Figure 1. Comparison of existing CLIP-based methods and our proposed CoPS. (a) Existing methods rely on human-designed prompts or static learnable tokens. (b) Our method proposes explicit state token synthesis and implicit class token sampling to synthesize visual-conditioned prompts.

capabilities by leveraging vision-language models (VLMs) pre-trained on large-scale image-text pairs. Building upon the success of CLIP [32], these methods design tailored prompts to adapt the model to downstream anomaly detection tasks. As illustrated in Fig. 1, each prompt can be decomposed into three components: context words, which describe the high-level imaging scene (e.g., "a photo of"); state words, which specify the normal or anomalous condition (e.g., "perfect"/"broken"); and class words, which capture the specific object category (e.g., "screw").

Existing methods can be broadly categorized into two types. As depicted in Fig. 1(a), prompt design methods such as AdaCLIP [4] rely on manually crafted template sets built from expert knowledge. While these prompts are intuitive and interpretable, their construction requires extensive manual effort through repeated trial-and-error. In contrast, prompt learning methods such as AnomalyCLIP [45] treat prompts as tunable parameters, replacing the context and state words with static learnable tokens and using a single class-agnostic label for all samples. However, this introduces two key limitations: *(i)* static learnable tokens fail to capture the continuous and diverse patterns of normal and anomalous states, limiting generalization to unseen categories; *(ii)* fixed textual labels offer overly sparse semantic information, making the model prone to overfitting within a narrow representation space.

To address these issues, we propose Conditional Prompt Synthesis (CoPS), a CLIP-based framework for ZSAD. We assume that context words can be shared across images using static learnable tokens, since high-level imaging context is largely invariant across categories. Meanwhile, the remaining prompt components are dynamically synthesized based on visual features, as shown in Fig. 1(b). Specifically, CoPS enhances prompt learning through two types of visual-conditioned token synthesis: *(i)* state words are injected with representative normal and anomaly prototypes, which are extracted under a center constraint from fine-grained local features, enabling better generalization through explicit state modeling; *(ii)* class words incorporate semantic global features sampled from a variational autoencoder (VAE) [22], enabling implicit label augmentation to improve prompt diversity. Leveraging the observation that the distance between query feature and its nearest prototype also approximates anomaly state, we introduce a distance-aware spatial attention mechanism to refine pixel-level text-image alignment. Additionally, we employ a global-local (glocal) similarity interaction that further strengthens image-level alignment.

Our contributions are summarized as follows:

- We propose CoPS, a novel framework built on CLIP to address the discretization of static learnable tokens and the sparsity of textual category labels, thereby improving ZSAD accuracy and generalization.
- We introduce two key modules to synthesize visual-conditioned prompts: an Explicit State Token Synthesis (ESTS) module for injecting normal and anomaly prototypes, and an Implicit Class Token Sampling (ICTS) module for fusing diverse semantic features.
- Integrated with a Spatially-Aware Glocal Alignment (SAGA) module, CoPS achieves state-of-the-art (SOTA) performance across 13 industrial and medical datasets under the ZSAD setting, improving classification and segmentation AUROC by 1.4% and 1.9%, respectively.

2. Related Work

2.1. Classical Anomaly Detection

In the classical AD setting, the model is trained solely on normal samples and detects anomalies within the same known category during inference. Specifically, reconstruction-based methods [11, 27, 40, 42] are widely used in classical AD, assuming the model accurately reconstructs normal samples but fails to reconstruct anomalies. Anomalies are then detected by analyzing the residuals between the input and its reconstruction. Embedding-based methods [23, 24, 33, 34] leverage pre-trained networks to extract features and compress normal features into a compact space. Anomalies are then detected by measuring the distance between test features and normal clusters. Since the aforementioned models cannot capture the anomalous distribution, synthesis-based methods [5–7, 20, 26, 37] augment training with synthetic anomalies to improve discrimination. However, these methods rely on specific datasets, limiting their applicability in real-world scenarios with distribution shifts between training and testing.

2.2. Zero-Shot Anomaly Detection

With the emergence of large pre-trained foundation models, ZSAD on unseen categories has become feasible due to CLIP's strong cross-modal generalization and image-text alignment capability [32]. Given the pivotal role of textual prompts in adapting CLIP to downstream tasks, ZSAD methods can be broadly categorized into two types.

Prompt design methods were introduced to model the relationship between normal and abnormal images using two sets of manually crafted textual templates. WinCLIP [17] is the first to apply CLIP to ZSAD. APRIL-GAN [8] and CLIP-AD [9] introduce trainable adapters for fine-grained alignment. AdaCLIP [4] further fine-tunes both text and vision encoders to enhance CLIP's ability to detect anomalies.

Prompt learning methods have recently gained attention, replacing fixed templates with learnable tokens to adaptively capture normal and anomalous semantics. AnomalyCLIP [45] builds on CoOp [44] by making the context words learnable. AdaptCLIP [12] adopts an alternating training strategy between textual and visual rep-

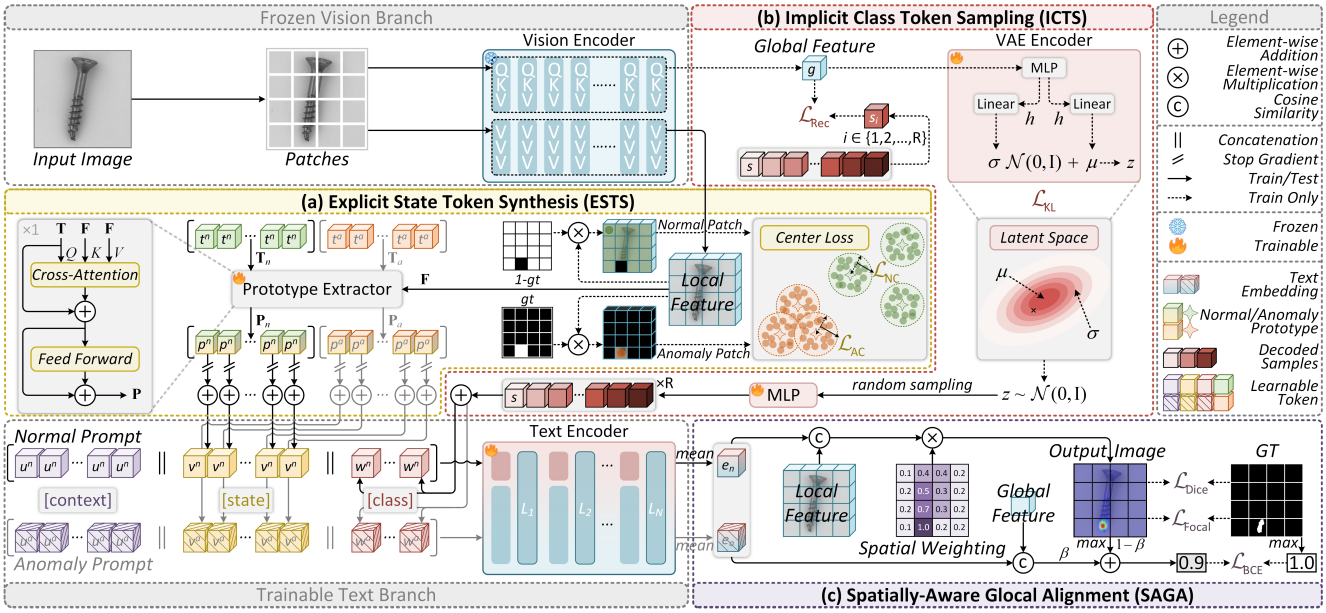


Figure 2. Schematic of CoPS. (a) ESTS extracts normal and anomaly prototypes from fine-grained local features and injects them into the state words. (b) ICTS leverages a VAE to model semantic global features and fuses decoded samples into the class words. (c) SAGA applies distance-aware spatial attention to align textual embeddings with global and local visual features.

resentations for more stable improvement. VCP-CLIP [30] extends CoCoOp [43] by embedding image features into class words, eliminating the need for category labels. Anomaly-OV [39] directly uses image features as class words and incorporates multimodal large language models (MLLMs) for anomaly reasoning. FAPrompt [46] synthesizes compound abnormality prompts by leveraging the mapped top-K abnormality-prior patch features. However, these prompts are weakly constrained and semantically limited, which hinders the generalization to unseen categories.

3. Method

3.1. Problem Definition

We follow the standard ZSAD setting, where the model is trained on seen categories $\mathcal{C}_s$ from an auxiliary training dataset and evaluated on disjoint unseen categories $\mathcal{C}_u$ (i.e., $\mathcal{C}_s \cap \mathcal{C}_u = \varnothing$) without any adaptation. Specifically, $\mathcal{C}_s$ are from an industrial dataset $\mathcal{X}_{\text{train}}$, while $\mathcal{C}_u$ are from industrial or medical datasets $\mathcal{X}_{\text{test}}$ with distinct visual patterns. The model outputs an image-level anomaly score and a pixel-level anomaly map for each image.

3.2. Overview

Fig. 2 illustrates the overall framework of CoPS, which consists of three key designs: *(i)* ESTS for state token synthesis, *(ii)* ICTS for class token sampling, and *(iii)* SAGA for glocal feature alignment. We employ Transformer-based vision and text encoders from the pre-trained CLIP [32] as the backbone. Given an input image $\mathbf{X} \in \mathbb{R}^{h\times w\times 3}$, the frozen vision encoder $\mathcal{V}_\rho$ first extracts semantic global feature $\mathbf{g} \in \mathbb{R}^C$ and fine-grained local feature $\mathbf{F} \in \mathbb{R}^{HW\times C}$, where $H = h/p$ and $W = w/p$ are the height and width for a patch size p, and C is the embedding dimension. Subsequently, ESTS introduces a prototype extractor $\mathcal{P}_\theta$ to generate M normal and anomaly prototypes $\mathbf{P}_n, \mathbf{P}_a \in \mathbb{R}^{M\times C}$ from local feature $\mathbf{F}$, which are assembled into the dynamic state tokens. ICTS then leverages a VAE $\mathcal{E}_\psi$ to parameterise the latent distribution of the global feature $\mathbf{g}$ and draws R decoded samples $\mathbf{S} \in \mathbb{R}^{R\times C}$, resulting in a dense set of class tokens. Normal and anomaly prompt sets $\{\tilde{\mathbf{G}}_i^n, \tilde{\mathbf{G}}_i^a\}_{i=1}^R$ are constructed by concatenating the static context tokens with the dynamic state tokens and sampled class tokens. To further enhance the image-text alignment, a learnable text encoder $\mathcal{T}_\omega$ is employed to map these prompts into the textual embeddings $\mathbf{e}_n, \mathbf{e}_a \in \mathbb{R}^C$. Finally, SAGA applies spatial attention to align $\mathbf{e}_n, \mathbf{e}_a$ with the global feature $\mathbf{g}$ and local feature $\mathbf{F}$, yielding an image-level anomaly score s_{cls} and a pixel-level anomaly map $\boldsymbol{\mathcal{S}}_{\text{seg}}$.

3.3. Learnable Dual-Prompt Construction

Following [31], we split text prompt into three components: context words, state words, and class words, as shown in the lower-left corner of Fig. 2. Recent work [45] builds upon the prompt learning strategy [44], by replacing fixed context templates with learnable tokens. However, they still manually craft a pair of state words (e.g., *"good"/"damaged"*) and fix the class words to class-agnostic text (e.g., *"ob-*

ject"), preventing either from being optimized and thereby limiting further gains in model generalization. Accordingly, we introduce learnable tokens to jointly optimize all three components, producing the initial normal and anomaly prompts $\mathbf{G}^n, \mathbf{G}^a \in \mathbb{R}^{L\times C}$ with sequence length L:

$$\begin{aligned}\mathbf{G}_n &= [\mathbf{u}_1^n]\cdots[\mathbf{u}_K^n] \parallel [\mathbf{v}_1^n]\cdots[\mathbf{v}_M^n] \parallel [\mathbf{w}_1^n]\cdots[\mathbf{w}_N^n] \\ \mathbf{G}_a &= [\mathbf{u}_1^a]\cdots[\mathbf{u}_K^a] \parallel [\mathbf{v}_1^a]\cdots[\mathbf{v}_M^a] \parallel [\mathbf{w}_1^a]\cdots[\mathbf{w}_N^a] \end{aligned} \quad (1)$$

where $\mathbf{u}_i^{n/a}, \mathbf{v}_i^{n/a}, \mathbf{w}_i^{n/a} \in \mathbb{R}^C$ are the learnable tokens in context, state, and class words, while K, M, and N are their corresponding lengths. The notation n/a serves as a placeholder for either the normal or anomalous variable.

3.4. Explicit State Token Synthesis (ESTS)

The normal and abnormal patterns of input images are continuous and diverse, making it difficult for fixed state words to generalize in zero-shot settings. To alleviate the overly discrete nature of the binary state text, the ESTS module maps the most representative normal and abnormal local patches into two prototypes and explicitly injects them into the prompt, enabling adaptive state modeling.

Since the original self-attention in CLIP builds relations among inconsistent semantic regions, we adopt consistent self-attention (i.e., V-VV) in the frozen vision encoder $\mathcal{V}_\rho$ to extract fine-grained local feature $\mathbf{F}$ without additional adaptation [25]. Fig. 2(a) introduces a prototype extractor $\mathcal{P}_\theta$ to compute the cross-attention between $\mathbf{F} \in \mathbb{R}^{HW\times C}$ and normal and anomalous learnable tokens $\mathbf{T}_{n/a} = \{\mathbf{t}_1^{n/a}, \mathbf{t}_2^{n/a}, \ldots, \mathbf{t}_M^{n/a} | \mathbf{t}_m^{n/a} \in \mathbb{R}^C\}$, yielding corresponding prototypes $\mathbf{P}_n, \mathbf{P}_a \in \mathbb{R}^{M\times C}$ for each state:

$$\begin{aligned}\mathbf{Q}_{n/a} &= \mathbf{T}_{n/a}\mathbf{W}^q, \mathbf{K}_{n/a} = \mathbf{F}\mathbf{W}^k, \mathbf{V}_{n/a} = \mathbf{F}\mathbf{W}^v \\ \mathbf{T}'_{n/a} &= \mathrm{Attention}(\mathbf{Q}_{n/a}, \mathbf{K}_{n/a}, \mathbf{V}_{n/a}) + \mathbf{T}_{n/a} \\ \mathbf{P}_{n/a} &= \mathrm{FFN}(\mathbf{T}'_{n/a}) + \mathbf{T}'_{n/a}\end{aligned} \quad (2)$$

where $\mathbf{W}^q, \mathbf{W}^k, \mathbf{W}^v \in \mathbb{R}^{C\times C}$ are the learnable projection matrices for queries $\mathbf{Q}_n, \mathbf{Q}_a \in \mathbb{R}^{M\times C}$, keys $\mathbf{K}_n, \mathbf{K}_a \in \mathbb{R}^{HW\times C}$, and values $\mathbf{V}_n, \mathbf{V}_a \in \mathbb{R}^{HW\times C}$.

Center loss. To ensure that these two prototypes correspond to the negative and positive states [28], we introduce normal center loss $\mathcal{L}_{\text{NC}}$ and abnormal center loss $\mathcal{L}_{\text{AC}}$ to maximize the cosine similarity between each token $\mathbf{f}_i \in \mathbb{R}^C$ of local feature $\mathbf{F}$ and its nearest prototype $\mathbf{p}_m \in \mathbf{P}$:

$$\begin{aligned} d_i^{n/a} &= \min_{m\in\{1,\ldots,M\}} \left(1 - \cos(\mathbf{f}_i, \mathbf{p}_m^{n/a})\right) \\ \mathcal{L}_{\text{ESTS}} &= \mathcal{L}_{\text{NC}} + \mathcal{L}_{\text{AC}} \\ &= \tfrac{1}{HW}\textstyle\sum_{i=1}^{HW}(d_i^n \cdot (1-y_i) + d_i^a \cdot y_i) \end{aligned} \quad (3)$$

where $\mathbf{y} = \{y_i\}_{i=1}^{HW}$ denotes the downsampled ground truth, with $y_i = 1$ for anomaly patches and $y_i = 0$ for normal ones. Fig. 3(a) illustrates the prototype matching process in ESTS, where $\mathbf{d}^{n/a} = \{d_i^{n/a}\}_{i=1}^{HW}$ is the distance between each query patch token and its 1-nearest neighbor (1-NN) prototype. As a result, the normal and anomaly prototypes $\mathbf{P}_{n/a} = \{\mathbf{p}_m^{n/a}\}_{m=1}^M$ are respectively injected into the learnable state words of dual prompts in Eq. (1).

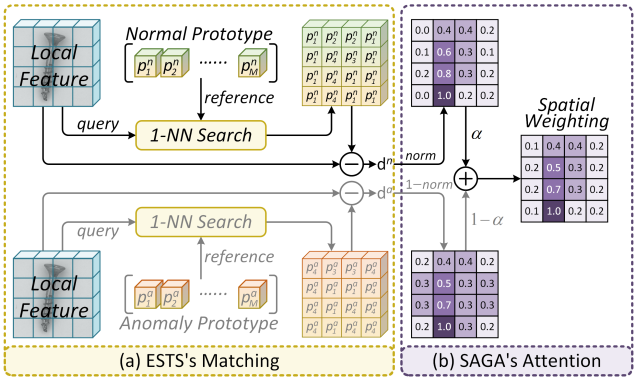


Figure 3. Pipeline of (a) ESTS's prototype matching and (b) SAGA's spatial attention.

3.5. Implicit Class Token Sampling (ICTS)

Existing methods represent class words for different categories using either multiple class-specific text labels [4] or a single class-agnostic text [45]. However, the learned semantic anchors in these models are overly sparse, making them prone to overfitting to a specific semantic subspace. Therefore, the ICTS module leverages a VAE $\mathcal{E}_\psi$ to implicitly fuse rich semantic information into the class words through distribution-consistent sampling.

The semantic global feature $\mathbf{g}$ is first extracted using the original self-attention mechanism (i.e., Q-KV) in the frozen vision encoder $\mathcal{V}_\rho$. As shown in Fig. 2(b), the VAE adopts a symmetric encoder–decoder architecture. The global feature $\mathbf{g} \in \mathbb{R}^C$ is fed into the VAE encoder $q_{\psi'}$ to produce the latent variable $\mathbf{z} \in \mathbb{R}^C$ via reparameterization from the estimated mean $\boldsymbol{\mu} \in \mathbb{R}^C$ and variance $\boldsymbol{\sigma} \in \mathbb{R}^C$:

$$\begin{aligned} &\mathbf{h} = \mathrm{MLP}_e(\mathbf{g}), \boldsymbol{\mu} = \mathbf{h}\mathbf{W}^\mu + \mathbf{b}^\mu, \log\boldsymbol{\sigma}^2 = \mathbf{h}\mathbf{W}^\sigma + \mathbf{b}^\sigma \\ &\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \end{aligned} \quad (4)$$

where $\mathbf{W}^\mu, \mathbf{W}^\sigma \in \mathbb{R}^{C\times C}$ and $\mathbf{b}^\mu, \mathbf{b}^\sigma \in \mathbb{R}^C$ are the learnable weights of the linear layers, and $\odot$ denotes the element-wise multiplication. The latent variable $\mathbf{z}$ is then fed into the decoder $p_{\psi''}$ to obtain the reconstructed global feature $\mathbf{s} \in \mathbb{R}^C$ (i.e., $\mathbf{s} = \mathrm{MLP}_d(\mathbf{z})$).

VAE loss. To synthesize diverse and robust class tokens, both the VAE encoder $q_{\psi'}$ and VAE decoder $p_{\psi''}$ are optimized by maximizing the evidence lower bound (ELBO):

$$\begin{aligned} \mathcal{L}_{\text{ICTS}} &= \mathcal{L}_{\text{Rec}} + \mathcal{L}_{\text{KL}} \\ &= \mathbb{E}_{q_{\psi'}(\mathbf{z}|\mathbf{g})}[-\log p_{\psi''}(\mathbf{g}|\mathbf{z})] + D_{\text{KL}}(q_{\psi'}(\mathbf{z}|\mathbf{g}) \parallel p(\mathbf{z})) \\ &= \|\mathbf{s}-\mathbf{g}\|^2 + \tfrac{1}{2}\textstyle\sum_{j=1}^{C}(\boldsymbol{\mu}_j^2 + \boldsymbol{\sigma}_j^2 - \log\boldsymbol{\sigma}_j^2 - 1) \end{aligned} \quad (5)$$

where $\mathcal{L}_{\text{Rec}}$ denotes the reconstruction loss and $\mathcal{L}_{\text{KL}}$ denotes the Kullback-Leibler (KL) divergence loss. The prior $p(\mathbf{z})$ is assumed to follow a multivariate standard normal distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$. We randomly sample R reconstructed global features $\mathbf{s}_i$ from the latent space via the VAE decoder to obtain a dense token set $\mathbf{S} = \{\mathbf{s}_i\}_{i=1}^{R}$. These are then fused into the learnable class words to generate the complete dual prompt sets $\{\tilde{\mathbf{G}}_i^n, \tilde{\mathbf{G}}_i^a\}_{i=1}^{R}$, where each pair of dual prompts $\tilde{\mathbf{G}}_i^n, \tilde{\mathbf{G}}_i^a \in \mathbb{R}^{L \times C}$ is constructed as:

$$\begin{aligned}\tilde{\mathbf{G}}_i^n &= [\mathbf{u}_1^n] \cdots [\mathbf{u}_K^n] \parallel [\mathbf{v}_1^n + \mathbf{p}_1^n] \cdots [\mathbf{v}_M^n + \mathbf{p}_M^n] \\ &\quad \parallel [\mathbf{w}_1^n + \mathbf{s}_i] \ \cdots \ [\mathbf{w}_N^n + \mathbf{s}_i] \\ \tilde{\mathbf{G}}_i^a &= [\mathbf{u}_1^a] \cdots [\mathbf{u}_K^a] \parallel [\mathbf{v}_1^a + \mathbf{p}_1^a] \cdots [\mathbf{v}_M^a + \mathbf{p}_M^a] \\ &\quad \parallel [\mathbf{w}_1^a + \mathbf{s}_i] \ \cdots \ [\mathbf{w}_N^a + \mathbf{s}_i] \end{aligned} \tag{6}$$

3.6. Spatially-Aware Glocal Alignment (SAGA)

Previous methods [8, 45] typically align textual embeddings independently with global and local features to obtain classification and segmentation results. However, this suffers from challenges in fine-grained and semantic text-image alignment. To enhance generalization and accuracy, the SAGA module proposes prototype-guided spatial attention and glocal similarity interaction.

Fig. 2(c) shows that the dual prompt sets $\{\tilde{\mathbf{G}}_i^n, \tilde{\mathbf{G}}_i^a\}_{i=1}^{R}$ are first fed into the text encoder $\mathcal{T}_\omega$, where the original prefix is replaced by learnable tokens. The normal and abnormal textual embeddings $\mathbf{e}_n, \mathbf{e}_a \in \mathbb{R}^C$ are then obtained by averaging over sampling dimension. To estimate the original likelihood of fine-grained patches and semantic image belonging to normal or anomalous categories, we compute the cosine similarity between textual embeddings $\mathbf{e}_n, \mathbf{e}_a$ and local feature $\mathbf{F}$ as well as global feature $\mathbf{g}$:

$$\begin{aligned}\boldsymbol{\mathcal{S}}_l^{n/a} &= \left\{ \frac{\exp(\cos(\mathbf{e}_{n/a}, \mathbf{f}_i)/\tau)}{\exp(\cos(\mathbf{e}_n, \mathbf{f}_i)/\tau) + \exp(\cos(\mathbf{e}_a, \mathbf{f}_i)/\tau)} \right\}_{i=1}^{HW} \\ s_g^{n/a} &= \frac{\exp(\cos(\mathbf{e}_{n/a}, \mathbf{g})/\tau)}{\exp(\cos(\mathbf{e}_n, \mathbf{g})/\tau) + \exp(\cos(\mathbf{e}_a, \mathbf{g})/\tau)} \end{aligned} \tag{7}$$

where τ is a temperature hyperparameter and $\mathbf{f}_i \in \mathbb{R}^C$ is the i-th patch token of $\mathbf{F}$, while $\boldsymbol{\mathcal{S}}_l^{n/a} \in \mathbb{R}^{HW}$ denotes the initial local similarity map and $s_g^{n/a} \in \mathbb{R}$ denotes the initial global similarity score.

Since a local token is more likely to be normal when its distance $\mathbf{d}^n$ to normal prototypes is small and its distance $\mathbf{d}^a$ to anomaly prototypes is large (and vice versa), we introduce prototype-based spatial attention to further enhance text-image alignment. As illustrated in Fig. 3(b), $\mathbf{d}^n$ and $\mathbf{d}^a$ are combined to compute the spatial weighting mask:

$$\mathbf{M}_{\text{mask}} = \alpha \cdot \frac{\mathbf{d}^n}{\|\mathbf{d}^n\|} + (1 - \alpha) \left(1 - \frac{\mathbf{d}^a}{\|\mathbf{d}^a\|}\right) \tag{8}$$

where distance coefficient α balances the influence of normal distance $\mathbf{d}^n$ and anomaly distance $\mathbf{d}^a$. The spatial weighting mask $\mathbf{M}_{\text{mask}} \in \mathbb{R}^{HW}$ is applied to refine the local similarity map, which is then aggregated to compute the refined global similarity score:

$$\begin{aligned}\hat{\boldsymbol{\mathcal{S}}}_l^{n/a} &= \boldsymbol{\mathcal{S}}_l^{n/a} \odot \mathbf{M}_{\text{mask}} \\ \hat{s}_g^{n/a} &= \beta \cdot s_g^{n/a} + (1 - \beta) \cdot \max(\hat{\boldsymbol{\mathcal{S}}}_l^{n/a}) \end{aligned} \tag{9}$$

where glocal coefficient β controls the trade-off between local and global scores.

Glocal loss. Following [45], we enhance the model's binary discriminative ability at both pixel and image levels by introducing a glocal loss defined as:

$$\begin{aligned}\mathcal{L}_{\text{SAGA}} &= \mathcal{L}_{\text{Dice}} + \mathcal{L}_{\text{Focal}} + \mathcal{L}_{\text{BCE}} \\ &= \text{Dice}(\tilde{\boldsymbol{\mathcal{S}}}_l^a, \mathbf{Y}) + \text{Dice}(\tilde{\boldsymbol{\mathcal{S}}}_l^n, \mathbf{1} - \mathbf{Y}) \\ &\quad + \text{Focal}([\tilde{\boldsymbol{\mathcal{S}}}_l^n, \tilde{\boldsymbol{\mathcal{S}}}_l^a], \mathbf{Y}) + \text{BCE}(\hat{s}_g^a, \max(\mathbf{Y})) \end{aligned} \tag{10}$$

where $\mathbf{Y} \in \mathbb{R}^{h \times w}$ denotes the full-size ground truth, and $\tilde{\boldsymbol{\mathcal{S}}}_l^{n/a} \in \mathbb{R}^{h \times w}$ is the reshaped and interpolated version of the refined local similarity map $\hat{\boldsymbol{\mathcal{S}}}_l^{n/a} \in \mathbb{R}^{HW}$.

3.7. Training and Inference

Since the prototype extractor $\mathcal{P}_\theta$ already enforces explicit prototype alignment through the center loss, the ESTS module is not optimized using SAGA's glocal loss. In contrast, the ICTS module implicitly samples from the global feature distribution and can be trained jointly with the VAE loss and glocal loss. Finally, the overall training objective is:

$$\mathcal{J} = \min_{\theta} \mathcal{L}_{\text{ESTS}} + \min_{\psi} \mathcal{L}_{\text{ICTS}} + \min_{\psi, \omega, \varphi} \mathcal{L}_{\text{SAGA}} \tag{11}$$

where θ, ψ, ω, and φ are the learnable parameters of the ESTS module, the ICTS module, the text encoder, and the initial dual prompts, respectively.

During inference, the vision encoder first extracts the global feature $\mathbf{g}$ and local feature $\mathbf{F}$ from input image $\mathbf{X}$. The state prototypes $\mathbf{P}$ are obtained from $\mathbf{F}$ via ESTS, and the class token sets $\mathbf{S}$ are sampled from the standard Gaussian prior via ICTS. These are then fused into the initial dual prompts to synthesize complete normal and anomaly prompt sets. Subsequently, the text encoder generates textual embeddings $\mathbf{e}$, which are aligned with $\mathbf{g}$ and $\mathbf{F}$ via SAGA. Finally, the image-level anomaly score s_{cls} is given by the refined global similarity score $\hat{s}_g^a$, and the pixel-level anomaly map $\boldsymbol{\mathcal{S}}_{\text{seg}}$ is obtained by applying Gaussian filtering to the refined local similarity map $\hat{\boldsymbol{\mathcal{S}}}_l^a$.

4. Experiments

4.1. Experimental Setup

Datasets. We evaluate the ZSAD performance of our proposed method on 13 publicly available datasets from industrial and medical domains. Specifically, we employ five

Table 1. Performance comparison of various SOTA methods on industrial and medical datasets under ZSAD setting, as measured by I-AUROC% / I-AP%. The best results are highlighted in bold, and the second-best results are underlined.

Domain ↓	Method →	Prompt Design				Prompt Learning		
	Dataset ↓	WinCLIP CVPR'23	APRIL-GAN CVPRw'23	CLIP-AD IJCAI'24	AdaCLIP ECCV'24	AnomalyCLIP ICLR'24	FAPrompt ICCV'25	**CoPS**
Industrial	MVTec-AD	91.8 / 95.1	86.1 / 93.5	89.8 / 95.3	92.0 / 96.4	91.5 / 96.2	91.9 / 95.9	**95.0 / 97.8**
	VisA	78.1 / 77.5	78.0 / 81.4	79.8 / 84.3	83.0 / 84.9	82.1 / 85.4	84.0 / 86.7	**85.4 / 88.0**
	BTAD	83.3 / 84.1	74.2 / 71.7	85.8 / 85.2	91.6 / 92.4	89.1 / 91.1	92.3 / 90.1	**93.6 / 94.9**
	MPDD	61.5 / 69.2	76.8 / **82.9**	71.6 / 76.3	76.4 / 80.4	73.7 / 76.5	76.5 / 80.3	**78.6** / 81.1
	DTD-Synth	95.0 / 97.9	83.9 / 93.6	91.5 / 96.8	92.8 / 97.0	94.5 / 97.7	94.9 / 97.6	**95.2 / 98.1**
Medical	HeadCT	83.7 / 81.6	89.3 / 89.6	93.8 / 92.2	93.4 / 92.2	95.3 / 95.2	**96.1** / 96.6	**96.1 / 97.1**
	BrainMRI	92.0 / 90.7	89.6 / 84.5	92.8 / 85.5	94.9 / 94.2	96.1 / 92.3	95.9 / 96.2	**97.4 / 97.6**
	Br35H	80.5 / 82.2	93.1 / 92.9	96.0 / 95.5	95.7 / 95.7	97.3 / 96.1	97.2 / 96.1	**98.7 / 98.5**
Average Performance		83.2 / 84.8	83.9 / 86.3	87.6 / 88.9	90.0 / 91.7	90.0 / 91.3	91.1 / 92.4	**92.5 / 94.1**

widely used industrial datasets: MVTec-AD [2], VisA [47], BTAD [29], MPDD [18], and DTD-Synthetic [1]. Additionally, we utilize eight medical datasets: HeadCT [34], BrainMRI [21], Br35H [14], ISIC [10], CVC-ColonDB [35], CVC-ClinicDB [3], Kvasir [19], and Endo [15]. Since the categories in VisA are disjoint from those in the other datasets, we use VisA as the auxiliary training set and evaluate on all remaining datasets. For VisA evaluation, the model is fine-tuned on the MVTec-AD dataset. All experiments are conducted under the same evaluation protocol to ensure a fair comparison. More detailed descriptions about the datasets can be found in Appendix Sec. A.

Evaluation metrics. To evaluate the ZSAD performance at both image and pixel levels, we employ the area under the receiver operating characteristic (AUROC) and average precision (AP) as evaluation metrics. AUROC and AP are denoted as I-AUROC/I-AP for the image-level classification and P-AUROC/P-AP for the pixel-level segmentation.

Implementation details. Following previous works [4, 45, 46], we adopt the publicly available CLIP (ViT-L/14@336px) pre-trained by OpenAI [32]. Input images are resized to 518×518, and the final layer of the vision encoder is used to extract global and local embeddings. For ESTS, the context token length K, state token length M, and class token length N are set to 6, 6, and 2, respectively. For ICTS, the sampling count R is set to 10. For SAGA, the distance coefficient α and glocal coefficient β are set to 0.3 and 0.9, respectively. CoPS is trained using the Adam optimizer for 10 epochs with an initial learning rate of 0.001 and a batch size of 8. All experiments are conducted on a system equipped with a single NVIDIA GeForce RTX 3090 GPU and an Intel Xeon Gold 6226R CPU. Additional implementation details can be found in Appendix Sec. B.

4.2. Comparison with State-of-the-Art

The proposed CoPS is compared with six SOTA methods on 13 industrial and medical datasets. Prompt design-based methods include: WinCLIP [17], APRIL-GAN [8], CLIP-AD [9], and AdaCLIP [4], while prompt learning-based methods include: AnomalyCLIP [45] and FAPrompt [46]. For fair comparison, all methods use the same backbone, input resolution, and evaluation protocol.

Quantitative results. As shown in Tab. 1, CoPS achieves the best or second-best performance on classification tasks across all industrial and medical datasets, with an average improvement of 1.4% in I-AUROC and 1.7% in I-AP. Benefiting from their tunable prompt representations, prompt learning-based methods generally outperform prompt design-based approaches on downstream tasks. CoPS further improves generalization by injecting constrained and sampled visual conditions into the synthesized prompts, leading to superior performance. Furthermore, Tab. 2 shows that CoPS also achieves state-of-the-art performance on segmentation tasks, with average improvements of 1.9% in P-AUROC and 4.2% in P-AP. Due to the extremely small defect regions in VisA and DTD-Synthetic, the AP metric is highly sensitive to over-detection. Although CoPS does not achieve the highest P-AP on these two datasets, it still surpasses our baseline AnomalyCLIP by 2.1% and 6.1%, respectively. Compared with AnomalyCLIP, CoPS benefits from spatial weighting, which further refines anomaly localization at the target regions. More quantitative results are presented in Appendix Sec. D.

Qualitative results. As shown in Fig. 4, we present the visualization of our CoPS and two representative ZSAD methods AdaCLIP [4] and AnomalyCLIP [45] on 15 categories from 10 datasets with pixel-level annotations. Fig. 4(a) shows that CoPS achieves precise localization on industrial datasets while reducing false positives on object foregrounds. In contrast, AdaCLIP produces numerous missed detections due to the high variance of anomaly scores and limited ability to capture large defect regions. Fig. 4(b) demonstrates that CoPS provides more comprehensive coverage of lesion regions on medical datasets.

Table 2. Performance comparison of various SOTA methods on industrial and medical datasets under ZSAD setting, as measured by P-AUROC% / P-AP%. The best results are highlighted in bold, and the second-best results are underlined.

Domain ↓	Method →	Prompt Design				Prompt Learning		
	Dataset ↓	WinCLIP CVPR'23	APRIL-GAN CVPRw'23	CLIP-AD IJCAI'24	AdaCLIP ECCV'24	AnomalyCLIP ICLR'24	FAPrompt ICCV'25	**CoPS**
Industrial	MVTec-AD	85.1 / 18.0	87.6 / 40.8	89.8 / 40.0	86.8 / 38.1	91.1 / 34.5	90.4 / 34.6	**93.4 / 41.9**
	VisA	79.6 / 5.00	94.2 / 25.7	95.0 / 26.3	95.1 / **29.2**	95.5 / 21.3	**95.7** / 21.4	**95.7** / 23.4
	BTAD	71.4 / 11.2	91.3 / 32.9	93.1 / **46.7**	87.7 / 36.6	93.3 / 42.0	93.7 / 40.4	**94.6** / 42.6
	MPDD	71.2 / 14.1	95.2 / 24.9	96.7 / 26.3	95.2 / 28.5	96.2 / 28.0	93.9 / 26.2	**97.5 / 30.9**
	DTD-Synth	82.5 / 11.6	96.6 / **67.3**	97.1 / 62.3	94.1 / 52.8	97.6 / 52.4	97.6 / 53.5	**98.4** / 58.5
Medical	ISIC	83.5 / 62.4	85.8 / 69.8	81.6 / 65.5	85.4 / 70.6	88.4 / 74.4	90.4 / 77.7	**93.8 / 85.6**
	ColonDB	64.8 / 14.3	78.4 / 23.2	80.3 / 23.7	79.3 / 26.2	81.9 / 31.3	85.2 / 36.7	**85.6 / 37.2**
	ClinicDB	70.7 / 19.4	86.0 / 38.8	85.8 / 39.0	84.3 / 36.0	85.9 / 42.2	85.5 / 45.1	**88.8 / 49.9**
	Kvasir	69.8 / 27.5	80.2 / 42.4	82.5 / 46.2	79.4 / 43.8	81.8 / 42.5	84.0 / 46.6	**85.8 / 51.5**
	Endo	68.2 / 23.8	84.1 / 47.9	85.6 / 51.7	84.0 / 44.8	86.3 / 50.4	88.4 / 55.8	**90.0 / 58.7**
Average Performance		74.7 / 20.7	87.9 / 41.4	88.8 / 42.8	87.1 / 40.7	89.8 / 41.9	90.5 / 43.8	**92.4 / 48.0**

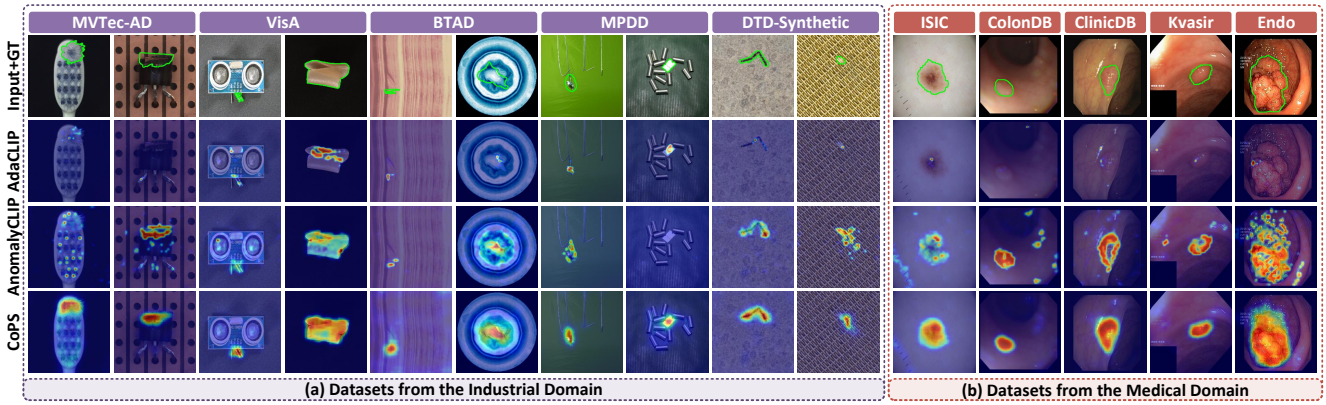


Figure 4. Qualitative comparison of CoPS against representative methods (AdaCLIP [4] and AnomalyCLIP [45]) across multiple datasets.

Meanwhile, AnomalyCLIP often yields over-detection because its normal and anomalous prompts lack continuity and diversity, leading to excessive activation in normal areas. These results indicate that our method synthesizes more effective prompts through prototype extraction and class sampling, which provide strong generalization for both industrial and medical ZSAD tasks. More qualitative results are presented in Appendix Sec. E.

4.3. Ablation Study

This section presents ablation studies to further assess the impact of each component and hyperparameters on CoPS.

Influence of different components. Tab. 3 reports ablation results of seven model variants against the complete model on four industrial and medical datasets to quantify the contribution of each component. All model variants retain learnable prompt tokens. Specifically, variant (A) serves as the baseline, similar to the AnomalyCLIP [45] framework. Variants (B-D) independently incorporate ESTS, ICTS, and SAGA into (A), each yielding performance gains. Notably, ESTS brings the largest improvement at both image and pixel levels, followed by SAGA and ICTS. Variants (E-G) are derived by removing SAGA, ICTS, and ESTS from the complete model, respectively. Both ICTS and SAGA further enhance classification and segmentation performance, while removing ESTS leads to the most significant drop by 1.4% in I-AUROC and 0.9% in P-AUROC. These results highlight the effectiveness of prototype extraction, class sampling, and glocal alignment.

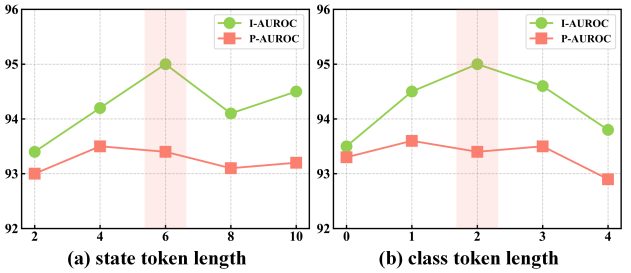


Figure 5. Performance ablation of state and class token length.

Table 3. Performance ablation of different components in CoPS on industrial and medical datasets under ZSAD setting, as measured by I-AUROC% / P-AUROC%. The best results are highlighted in bold, and the second-best results are underlined.

Module ↓	Model Variants							Ours
	(A)	(B)	(C)	(D)	(E)	(F)	(G)	
ESTS	×	✔	×	×	✔	✔	×	✔
ICTS	×	×	✔	×	✔	×	✔	✔
SAGA	×	×	×	✔	×	✔	✔	✔
MVTec-AD	91.1 / 92.0	93.2 / 93.0	91.7 / 92.0	92.4 / 92.2	94.5 / 93.3	93.5 / 93.3	92.6 / 92.2	**95.0 / 93.4**
BTAD	91.6 / 92.4	93.0 / 93.7	92.0 / 92.9	91.9 / 94.0	93.4 / 93.8	93.3 / 94.4	92.4 / 94.2	**93.6 / 94.6**
HeadCT	94.2 / -	94.8 / -	94.3 / -	94.6 / -	95.0 / -	95.5 / -	95.6 / -	**96.1 / -**
Endo	- / 88.5	- / 89.0	- / 88.6	- / 88.9	- / 89.3	- / 89.7	- / 89.1	**- / 90.0**
Average	92.3 / 91.0	93.7 / 91.9	92.7 / 91.2	93.0 / 91.7	94.3 / 92.1	94.1 / 92.5	93.5 / 91.8	**94.9 / 92.7**

Table 4. Performance ablation of sampling count R. The selected sampling count is underlined.

R	I-AUROC	I-AP	P-AUROC	P-AP	1/FPS
0	93.5%	97.2%	93.3%	41.8%	140ms
1	94.5%	97.6%	93.5%	41.9%	144ms
5	94.9%	97.8%	93.4%	42.0%	154ms
10	95.0%	97.8%	93.4%	41.9%	168ms
15	94.9%	97.7%	93.4%	41.8%	179ms
20	94.9%	97.7%	93.5%	41.9%	185ms

Influence of learnable token length. Fig. 5(a) illustrates the effect of varying the state token length M, which corresponds to the number of normal and abnormal prototypes in ESTS. Performance increases from $M = 2$ to $M = 6$, suggesting that a larger prototype set better captures state diversity. However, further increasing M to 8 and 10 results in a slight performance drop, likely caused by overfitting. Fig. 5(b) shows the effect of varying class token length N in ICTS. Performance increases from $N = 0$ to $N = 2$, suggesting that small-scale class sampling enhances semantic diversity. Similarly, performance drops when N exceeds 2, indicating diminishing returns from additional class words.

Influence of sampling count. Tab. 4 presents the ablation of sampling count R in ICTS. Increasing R from 0 to 10 leads to consistent gains at the image level and relatively minor gains at the pixel level, indicating that class sampling enriches prompt diversity. Since performance saturates while inference time increases beyond $R = 10$, this value offers a balanced trade-off between accuracy and efficiency. We select $R = 10$ as the default setting.

Influence of weight coefficients. As shown in Fig. 6(a), increasing the distance coefficient α in SAGA first improves performance and then leads to a decline. We select $\alpha = 0.3$ to balance the contributions of normal and abnormal prototype distances in the spatial weighting mask. As shown in Fig. 6(b), increasing the glocal coefficient β consistently improves both I-AUROC and I-AP, with peak performance at $\beta = 0.9$. This trend highlights the effectiveness of glocal alignment in integrating global and local semantics for anomaly detection. However, performance drops notably at $\beta = 1.0$, indicating that relying solely on global information without local score hinders generalization. More ablation study can be found in Appendix Sec. C.

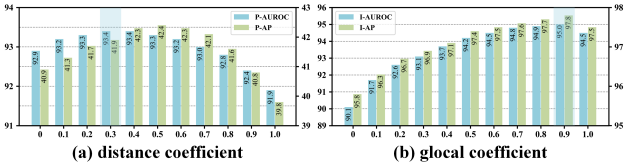


Figure 6. Performance ablation of distance and glocal coefficient.

5. Conclusion

In this work, we propose CoPS, a conditional prompt synthesis framework for ZSAD. By explicitly injecting state prototypes and implicitly fusing class semantics, CoPS synthesizes dynamic prompts conditioned on visual features, addressing the limitations of discrete state tokens and sparse class labels. Furthermore, our spatially-aware glocal alignment enhances generalization and accuracy. These designs enable CoPS to achieve SOTA zero-shot anomaly classification and segmentation on 13 industrial and medical datasets.

Limitations & Future Work. Although CoPS shows strong performance in detecting structural anomalies across diverse industrial and medical datasets, it remains limited in handling anomalies that require high-level semantic reasoning. This limitation stems from the lack of deep understanding of object relations and functional semantics within current framework. To this end, future work will integrate MLLMs to enhance CoPS's logical anomaly detection capability and improve generalization to complex scenarios.

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